7.7 Hyperbolic Functions

The hyperbolic functions are formed by taking combinations of the two exponential functions e^x and e^{-x}.

Recall that we say a function $f(x)$ is an **even** function if $f(-x) = f(x)$,

and that it is an **odd** function if $f(-x) = -f(x)$.

Given any function $f(x)$, we can write it as a sum of even and odd functions:

$$f(x) = \underbrace{\frac{f(x) + f(-x)}{2}}_{\text{even}} + \underbrace{\frac{f(x) - f(-x)}{2}}_{\text{odd}}.$$

So if we let $f(x) = e^x$, we get $e^x = \underbrace{\frac{e^x + e^{-x}}{2}}_{\text{even}} + \underbrace{\frac{e^x - e^{-x}}{2}}_{\text{odd}}$.

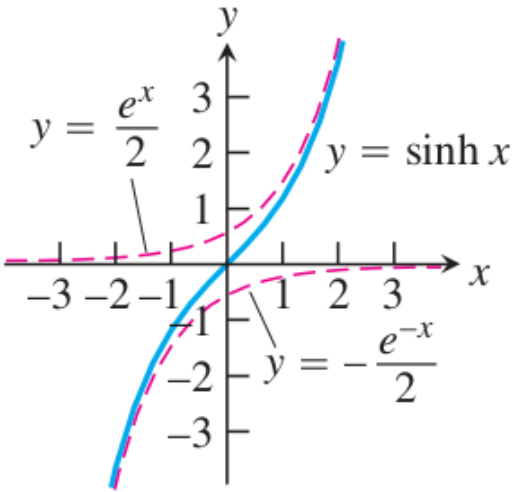
$\cosh x$ $\qquad$ $\sinh x$

Definitions

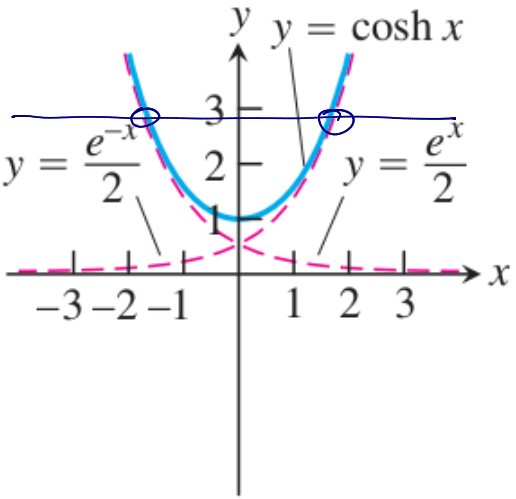
$$e^x = \frac{e^x + e^{-x}}{2} + \frac{e^x - e^{-x}}{2}$$

Here we define

$$\sinh x = \frac{e^x - e^{-x}}{2} \text{ (odd)} \quad \& \quad \cosh x = \frac{e^x + e^{-x}}{2} \text{ (even).}$$

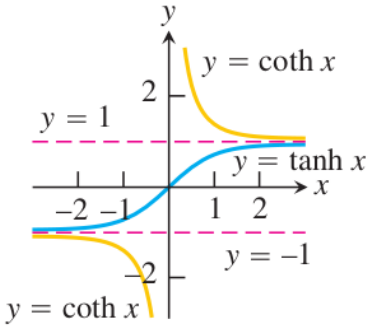


(a)

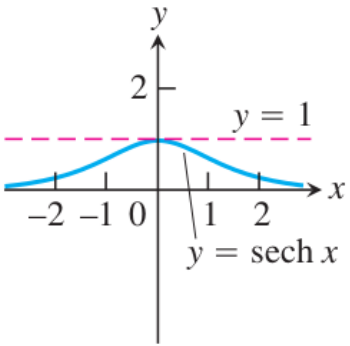


(b)

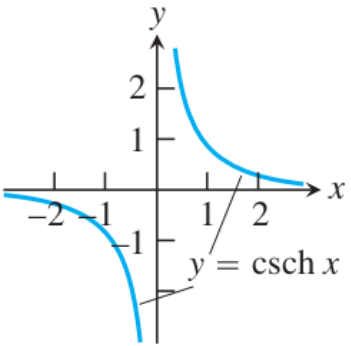
The graphs of the other hyperbolic functions.



(c)



(d)



(e)

$\tanh x = \dfrac{\sinh x}{\cosh x}$

$= \dfrac{e^x - e^{-x}}{e^x + e^{-x}}$

$\operatorname{sech} x = \dfrac{1}{\cosh x}$

$= \dfrac{2}{e^x + e^{-x}}$

$\operatorname{csch} x = \dfrac{1}{\sinh x}$

$= \dfrac{2}{e^x - e^{-x}}$

$\coth x = \dfrac{\cosh x}{\sinh x} = \dfrac{e^x + e^{-x}}{e^x - e^{-x}}$

Some identities for hyperbolic functions:

Show that

1. $\cosh^2 x - \sinh^2 x = 1$
2. $\sinh 2x = 2 \sinh x \cosh x$
3. $\cosh 2x = \cosh^2 x + \sinh^2 x$

1. $\cosh^2 x - \sinh^2 x = \left(\frac{e^x + e^{-x}}{2}\right)^2 - \left(\frac{e^x - e^{-x}}{2}\right)^2$

$= \frac{e^{2x} + 2 + e^{-2x}}{4} - \frac{e^{2x} - 2 + e^{-2x}}{4} = \frac{4}{4} = 1.$

2. $2 \sinh x \cosh x = \cancel{2}\left(\frac{e^x - e^{-x}}{\cancel{2}}\right)\left(\frac{e^x + e^{-x}}{2}\right) = \frac{e^{2x} - e^{-2x}}{2}$

3. $\cosh^2 x + \sinh^2 x = \left(\frac{e^x + e^{-x}}{2}\right) + \left(\frac{e^x - e^{-x}}{2}\right)^2 = \sinh 2x.$

$= \frac{e^{2x} + e^{-2x}}{2} = \cosh 2x$

Identities for hyperbolic functions:

- $\cosh^2 x - \sinh^2 x = 1$
- $\sinh 2x = 2 \sinh x \cosh x$
- $\cosh 2x = \cosh^2 x + \sinh^2 x$
- $\cosh^2 x = \frac{\cosh 2x + 1}{2}$
- $\sinh^2 x = \frac{\cosh 2x - 1}{2}$
- $\tanh^2 x = 1 - \operatorname{sech}^2 x$
- $\coth^2 x = 1 + \operatorname{csch}^2 x$

identities involving trigo. functions

$\cos^2\theta + \sin^2\theta = 1$

$\sin 2\theta = 2 \sin\theta \cos\theta$

$\cos 2\theta = \cos^2\theta - \sin^2\theta$

$\cos^2\theta = \frac{\cos 2\theta + 1}{2}$

$\sin^2\theta = \frac{1 - \cos 2\theta}{2}$

$\tan^2\theta = \sec^2\theta - 1$

$\cot^2\theta = \csc^2\theta - 1$

Derivatives of hyperbolic functions

- $\frac{d}{dx}(\sinh u) = \cosh u \frac{du}{dx}$

$$\frac{d}{dx}(\sinh x) = \frac{d}{dx}\left(\frac{e^x - e^{-x}}{2}\right)$$

$$= \frac{e^x - (-e^{-x})}{2} = \frac{e^x + e^{-x}}{2} = \cosh x$$

Chain Rule: $\frac{d}{dx}(\sinh u) = \cosh u \frac{du}{dx}$

- $\frac{d}{dx}(\text{csch} u) = -\text{csch} u \coth u \frac{du}{dx}$

$$\frac{d}{dx}(\text{csch} x) = \frac{d}{dx}\left(\frac{1}{\sinh x}\right)$$

$$= -\frac{1}{(\sinh x)^2}\frac{d}{dx}(\sinh x) = -\frac{1}{(\sinh x)^2}\cosh x$$

$$= -\left(\frac{\cosh x}{\sinh x}\right)\left(\frac{1}{\sinh x}\right)$$

$$= -\coth x \, \text{csch} x$$

$\frac{d}{dx}\cosh x = \sinh x$

$\frac{d}{dx}\tanh x = \text{sech}^2 x$

$\frac{d}{dx}\coth x = -\text{csch}^2 x$

$\frac{d}{dx}\text{sech} x = -\text{sech} x \tanh x$

Example:

a. $\frac{d}{dx}(\cosh\sqrt{1+x^2}) = \sinh\sqrt{1+x^2}\,\frac{d}{dx}\sqrt{1+x^2}$

$u = \sqrt{1+x^2}$ $= \sinh\sqrt{1+x^2}\,\frac{2x}{2\sqrt{1+x^2}} = \frac{\sinh\sqrt{1+x^2}}{\sqrt{1+x^2}}$

b. $\frac{d}{dx}(e^x \sinh x) = e^x \sinh x + e^x \cosh x$

$= e^x(\sinh x + \cosh x) = e^x e^x = \underline{\underline{e^{2x}}}$.

c. $\frac{d}{dx}[(x^2-1)\operatorname{csch}(\ln x)] = \frac{d}{dx}\left[(x^2-1)\frac{2}{e^{\ln x} - e^{-\ln x}}\right]$

$e^{-\ln x} = e^{\ln\frac{1}{x}} = \frac{1}{x}$

$= \frac{d}{dx}\left[(x^2-1)\frac{2}{x - \frac{1}{x}}\right]$

$= \frac{d}{dx}\left[(x^2-1)\frac{2x}{(x^2-1)}\right] = \frac{d}{dx}(2x) = \underline{\underline{2}}$

d. $\int \coth 5x\,dx = \int \frac{\cosh 5x}{\sinh 5x}\,dx = \int \frac{1}{u}\cdot\frac{du}{5}$

$u = \sinh 5x$

$du = 5\cosh 5x\,dx$

$= \frac{1}{5}\ln|u| + C$

$= \frac{1}{5}\ln|\sinh 5x| + C$

e. $\int_0^{\ln 2} 4e^x \sinh x\,dx = \int_0^{\ln 2} 4e^x \frac{e^x - e^{-x}}{2}\,dx$

$= 2\int_0^{\ln 2}(e^{2x} - 1)\,dx$

$= 2\left[\frac{e^{2x}}{2} - x\right]_0^{\ln 2}$

$e^{2\ln 2} = 4$

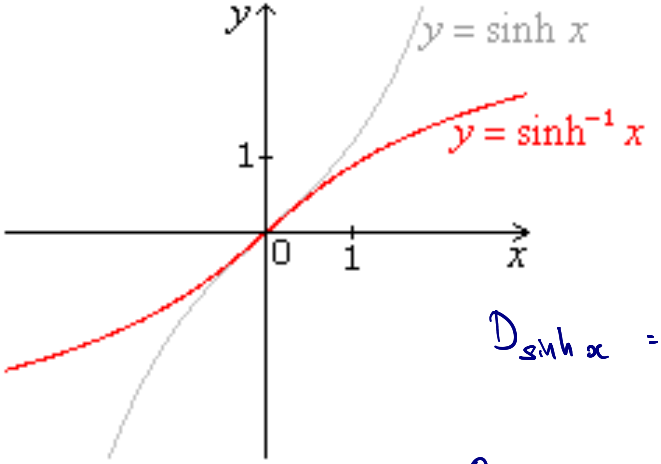
$= 2\left[\left(\frac{e^{2\ln 2}}{2} - \ln 2\right) - \frac{1}{2}\right]$

$= 4 - 2\ln 2 - 1 = 3 - 2\ln 2$

Inverse Hyperbolic Functions

The inverses of the six basic hyperbolic functions are very useful in integration.

$f(x) = \sinh x$ is one-to-one. $\Rightarrow f^{-1}(x)$, we denote the inverse by $\sinh^{-1} x$.



$D_{\sinh x} = R_{\sinh^{-1} x} = \mathbb{R} = (-\infty, \infty)$

$R_{\sinh x} = D_{\sinh^{-1} x} = \mathbb{R} = (-\infty, \infty)$.

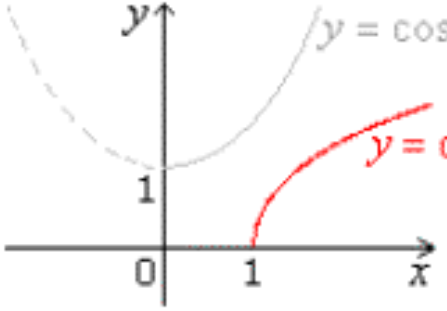
$f(x) = \cosh x$ is not one-to-one, but the restricted function $f(x) = \cosh x,\ x \in [0, \infty)$ is one-to-one.

$\cosh^{-1} x$.

Restricted

$D_{\cosh x} = [0, \infty)$

$\Rightarrow R_{\cosh^{-1} x} = [0, \infty)$



$R_{\cosh x} = [1, \infty)$

$D_{\cosh^{-1} x} = [1, \infty)$

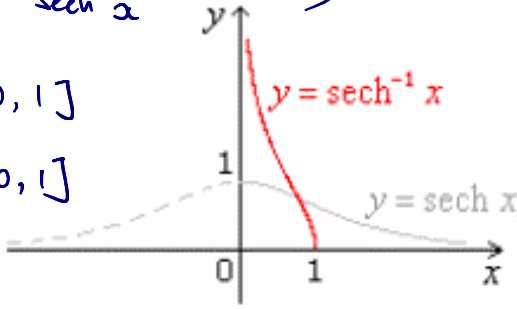
$f(x) = \text{sech } x$ is not one-to-one, but the restricted function $f(x) = \text{sech } x,\ x \in [0, \infty)$ is one-to-one.

$D_{\text{sech } x} = R_{\text{sech}^{-1} x} = [0, \infty)$

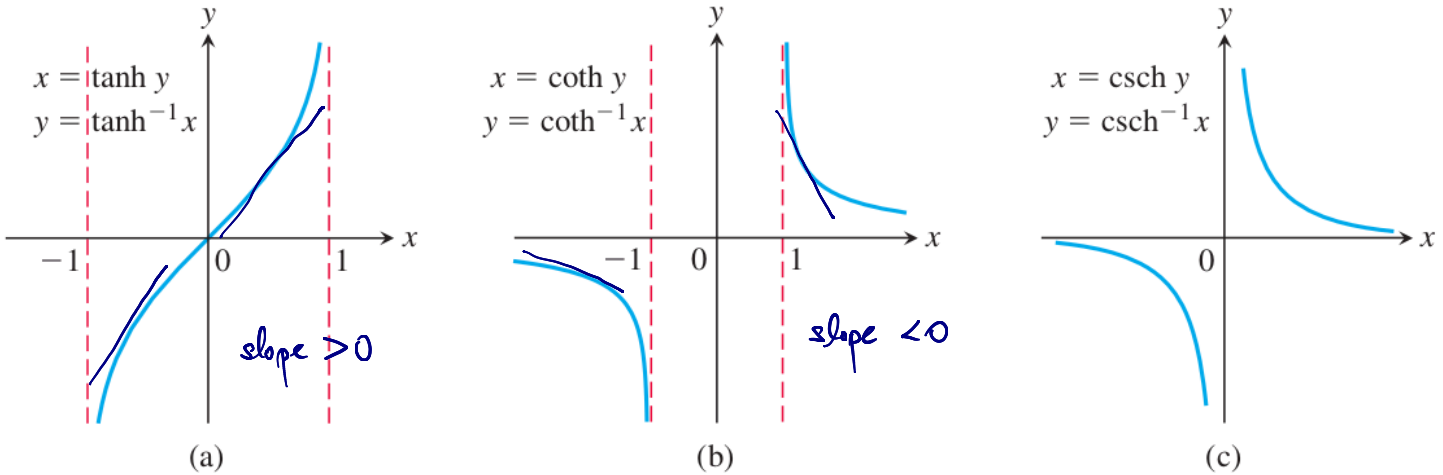
$f^{-1}(x) = \text{sech}^{-1} x$.

$R_{\text{sech } x} = (0, 1]$

$\therefore D_{\text{sech}^{-1} x} = (0, 1]$



The hyperbolic tangent, cotangent, and cosecant are one-to-one on their domains and therefore have inverses. ~~(Page 441, Fig. 7.33)~~



Remark: 1. The number $y = \sinh^{-1} x$ is the number in $(-\infty, \infty)$ for which $\sinh y = x$.

2. Caution that $\sinh^{-1} x \neq \frac{1}{\sinh x}$.

Identities for Inverse Hyperbolic Functions

- $\operatorname{sech}^{-1} x = \cosh^{-1} \frac{1}{x}, \; x \in (0, 1]. \; (*)$
- $\operatorname{csch}^{-1} x = \sinh^{-1} \frac{1}{x}, \; x \in (-\infty, 0) \cup (0, \infty).$
- $\coth^{-1} x = \tanh^{-1} \frac{1}{x}, \; x \in (-\infty, 1) \cup (1, \infty).$ $\Rightarrow \coth x = \frac{1}{\tanh x}$.

Proof of (*)

Observe that $\operatorname{sech}\left(\cosh^{-1}\left(\frac{1}{x}\right)\right) = \frac{1}{\cosh\left(\cosh^{-1}\left(\frac{1}{x}\right)\right)}$

$= \frac{1}{1/x} = x$

Since $\operatorname{sech}\left(\cosh^{-1}\left(\frac{1}{x}\right)\right) = x$. $\operatorname{sech} y = x$

$\Rightarrow \operatorname{sech}^{-1} x = \cosh^{-1}\left(\frac{1}{x}\right)$ $\Rightarrow \operatorname{sech}^{-1} x = y$

Derivatives of inverse hyperbolic functions

$\frac{d}{dx}(\cosh^{-1}x) = \frac{1}{\sqrt{x^2-1}}, \; x > 1$

$f(x) = \cosh x \Rightarrow f'(x) = \sinh x$

$\Rightarrow \frac{d}{dx}(\cosh^{-1}x) = (f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$

$= \frac{1}{\sinh(\cosh^{-1}x)}$

Recall that $\cosh^2\alpha - \sinh^2\alpha = 1$, we have $\sinh\alpha = \pm\sqrt{\cosh^2\alpha - 1}$

$\Rightarrow \frac{d}{dx}(\cosh^{-1}x) = \frac{1}{\pm\sqrt{\cosh^2(\cosh^{-1}x)-1}} = \frac{1}{\pm\sqrt{x^2-1}}$

$= \frac{1}{\sqrt{x^2-1}}$ (since $\cosh^{-1}x$ has positive slope, as observed from the graph.)

Apply CR: $\frac{d}{dx}(\cosh^{-1}u) = \frac{1}{\sqrt{u^2-1}}\frac{du}{dx}$, $u > 1$.

Show that $\frac{d}{dx}(\sinh^{-1}x) = \frac{1}{\sqrt{1+x^2}}\ \forall\, x \in \mathbb{R}$.

Show that $\frac{d}{dx}(\tanh^{-1}x) = \frac{1}{1-x^2},\ |x| < 1.$ always +

Show that $\frac{d}{dx}(\coth^{-1}x) = \frac{1}{1-x^2},\ |x| > 1.$ always −

(1) Let $y = \sinh^{-1} x$

$\Rightarrow \sinh y = x$ (dif wrt x) $\Rightarrow \cosh y \frac{dy}{dx} = 1$

$\Rightarrow \frac{dy}{dx} = \frac{1}{\cosh y} = \frac{1}{\sqrt{1+\sinh^2 y}}$ ($\cosh^2 y - \sinh^2 y = 1$)

$= \frac{1}{\sqrt{1+x^2}}$

Example:

a. $\frac{d}{dx}(\tanh^{-1}(\sin 2x)) = \frac{1}{1-(\sin 2x)^2} \frac{d}{dx}(\sin 2x)$

$u = \sin 2x$

$= \frac{1}{1-\sin^2 2x} \cdot 2\cos 2x$

$= \frac{1}{\cos^2 2x} \cdot 2\cos 2x = \frac{2}{\cos 2x} = 2\sec 2x$

b. $\frac{d}{dx}(\cosh^{-1}(x^2+1)) = \frac{1}{\sqrt{(x^2+1)^2-1}} \frac{d}{dx}(x^2+1)$

$u = x^2+1$

$= \frac{1}{\sqrt{x^4+2x^2}} \cdot 2x$

$= \frac{2x}{\sqrt{x^4+2x^2}}$

Derivatives of inverse hyperbolic functions

1. $\frac{d(\sinh^{-1}u)}{dx} = \frac{1}{\sqrt{1+u^2}}\frac{du}{dx}, \ u \in \mathbb{R}$

2. $\frac{d(\cosh^{-1}u)}{dx} = \frac{1}{\sqrt{u^2-1}}\frac{du}{dx}, \ u > 1$

3. $\frac{d(\tanh^{-1}u)}{dx} = \frac{1}{1-u^2}\frac{du}{dx}, \ |u| < 1$

4. $\frac{d(\coth^{-1}u)}{dx} = \frac{1}{1-u^2}\frac{du}{dx}, \ |u| > 1$

5. $\frac{d(sech^{-1}u)}{dx} = -\frac{1}{u\sqrt{1-u^2}}\frac{du}{dx}, \ 0 < u < 1$

6. $\frac{d(csch^{-1}u)}{dx} = -\frac{1}{|u|\sqrt{1+u^2}}\frac{du}{dx}, \ u \neq 0$

Integrals leading to inverse hyperbolic functions

$\frac{d}{dx}(\sinh^{-1} u) = \frac{1}{\sqrt{1+u^2}} \frac{du}{dx}$.

1. $\int \frac{du}{\sqrt{a^2+u^2}} = \sinh^{-1}(\frac{u}{a}) + C,\ a > 0.$

$\Rightarrow \int \frac{1}{\sqrt{1+u^2}} du = \sinh^{-1} u + C$

2. $\int \frac{du}{\sqrt{u^2-a^2}} = \cosh^{-1}(\frac{u}{a}) + C,\ u > a > 0.$

$u = \frac{w}{a}$, $du = \frac{1}{a} dw$

$\Rightarrow \int \frac{1}{\sqrt{1+\frac{w^2}{a^2}}} \frac{dw}{a} = \sinh^{-1}\left(\frac{w}{a}\right)$

3. $\int \frac{du}{a^2-u^2} = \frac{1}{a}\tanh^{-1}(\frac{u}{a}) + C, \text{ if } u^2 < a^2.$

4. $\int \frac{du}{a^2-u^2} = \frac{1}{a}\coth^{-1}(\frac{u}{a}) + C, \text{ if } u^2 > a^2.$

$\Rightarrow \int \frac{1}{\sqrt{a^2+w^2}} dw = \sinh^{-1}\left(\frac{w}{a}\right)$

5. $\int \frac{du}{u\sqrt{a^2-u^2}} = -\frac{1}{a}\text{sech}^{-1}(\frac{u}{a}) + C, 0 < u < a.$

6. $\int \frac{du}{u\sqrt{a^2+u^2}} = -\frac{1}{a}\text{csch}^{-1}|\frac{u}{a}| + C, u \neq 0 \text{and } a > 0.$

Example: $\int \frac{2dx}{\sqrt{3+4x^2}} =$ $\int \frac{du}{\sqrt{a^2+u^2}} = \sinh^{-1}\left(\frac{u}{a}\right) + C$

$\sqrt{a^2+u^2}$

$u = 2x \Rightarrow du = 2\,dx$

$a = \sqrt{3}$

$= \sinh^{-1}\left(\frac{2x}{\sqrt{3}}\right) + C$

- $\int \frac{3\,dx}{\sqrt{9x^2+6x}} = \int \frac{3}{\sqrt{(3x+1)^2-1}}\,dx$

$9x^2+6x = (3x+1)^2 - 1$

$u = 3x+1 \quad du = 3\,dx$

$a = 1$

$= \int \frac{du}{\sqrt{u^2-a^2}}$

$= \cosh^{-1}\left(\frac{u}{a}\right) + C$

$= \cosh^{-1}(3x+1) + C$